WebDrop

MOBILE-FIRST NEARBY TRANSFER PROTOTYPE

# WebDrop

## Demo Guide

VERSION 1.0.10

A visual and technical guide to the WebDrop 1.0.10 prototype. This document explains the product in plain language, shows every approved interface state, and records the boundary between the working local demo and the future production services.

English edition

01 / PRODUCT

# Overview

WebDrop is designed to make sending a file to someone nearby feel as immediate as tapping their profile. The home screen turns nearby devices into calm orbital avatars. A user chooses one person, confirms the connection with a deliberate swipe, and then gets separate controls for sending, receiving, chatting, or disconnecting.

### What the user sees

Four rings organize nearby candidates without turning the screen into a list. The current device stays at the center. Other profiles remain static while their positions move around the rings, keeping identity readable and motion restrained.

### What happens in the background

The prototype separates coordination from file movement. Signaling only exchanges small messages such as invitations, verification telemetry, WebRTC offers, answers, and ICE candidates. File bytes are reserved for an encrypted WebRTC DataChannel.

### Why verification exists

A nearby name alone is not enough proof that the intended person is holding that device. WebDrop therefore scaffolds a proximity ceremony using QR confirmation or optional sound, motion, bump, and tilt clues. Permissions and device support can always force the safer QR fallback.

### What this build proves

Version 1.0.10 proves the mobile-first interaction, responsive orbit layout, bilingual interface, local connection simulation, transfer states, storage strategy, and replaceable service boundaries. It does not yet claim production internet signaling or managed TURN capacity.

02 / EXPERIENCE

# How a transfer works

The following sequence describes the intended production path in simple terms. The current build locally simulates the service-dependent steps while keeping the same interfaces the backend will later implement.

1

### Discover

A signaling service announces small presence records so nearby or eligible devices can appear on the orbit. No file payload is sent.

2

### Choose and verify

The sender taps one candidate and swipes to confirm. QR or optional sensor evidence helps both sides prove they mean the same nearby peer.

3

### Negotiate WebRTC

The signaling adapter carries offers, answers, and ICE candidates. WebRTC then tries a direct path and can later use TURN as a relay.

4

### Move file chunks

The file is divided into modest chunks, around 64 KiB by default. Backpressure keeps the DataChannel buffer from growing without limit.

5

### Store safely

A worker writes received chunks to OPFS when available, then IndexedDB. Memory is only appropriate for small fallback transfers.

6

### Open or export

Private browser storage and a user-visible download are separate operations. The receive sheet exposes completed files only after the local write and integrity bookkeeping finish.

03 / SYSTEM

# Technology and architecture

The static application is intentionally split into small, replaceable parts. The browser experience can ship independently while production signaling, TURN credentials, and server policy are developed next.

INDEX.HTML

### Semantic shell

Contains accessible landmarks and component mounting points only. Business behavior stays out of the document structure.

CSS MODULES

### Visual system

Tokens, orbit geometry, controls, sheets, transfer states, print rules, and responsive behavior are separated by responsibility.

VANILLA JS MODULES

### Application logic

Bootstrap, state transitions, UI rendering, capabilities, proximity, signaling, WebRTC, transfer, storage, and utilities remain independent.

SIGNALING ADAPTERS

### Replaceable coordination

The mock adapter powers this demo. A WebSocket adapter will later carry presence and RTC metadata without carrying file payloads.

WEBRTC DATACHANNEL

### Encrypted peer transport

Files travel as buffered chunks between peers. Candidate statistics can classify a successful route as direct or TURN-relayed.

WORKERS AND STORAGE

### Large-file safety

Workers keep hashing, chunk bookkeeping, and local writes away from the main UI thread. OPFS leads the capability ladder.

04 / BOUNDARY

# Current scope and limitations

The interface is functional as a local demonstration, but several network and browser realities still determine production success.

### Mock signaling

Invites and RTC metadata are simulated locally. The production WSS service, authentication, presence policy, and abuse controls remain pending.

### TURN is not provisioned

Some networks cannot establish a direct path. Managed TURN credentials, relay quotas, and a conservative relayed-file size cap are future work.

### Permissions vary

Microphone and motion APIs require secure contexts, user gestures, and permission. QR remains the dependable verification fallback.

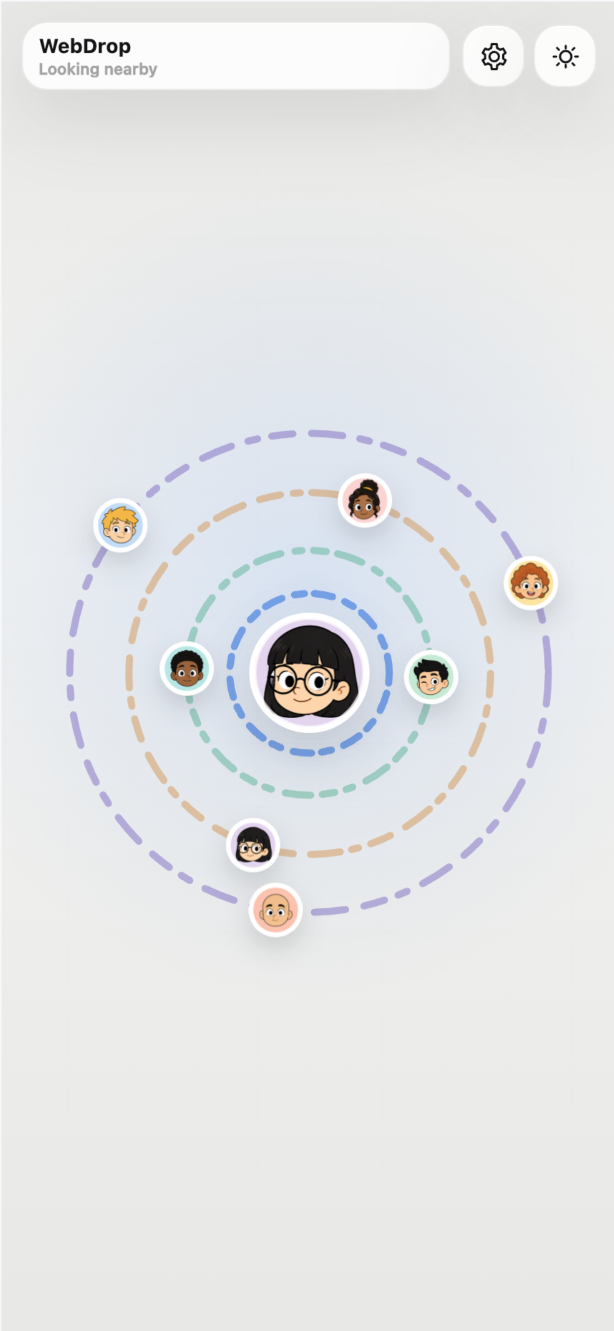
### Storage is capability-based

OPFS is preferred, IndexedDB is the fallback, and memory is limited to small files. Export can still fail independently of private storage.

05 / UI CATALOG  01

# Interface guide

Approved WebDrop 1.0.10 UI inventory




## Light home screen

The main nearby-device surface in light mode, with the WebDrop status, four orbits, a gently animated self icon, and static orbit peers.

# Interface guide

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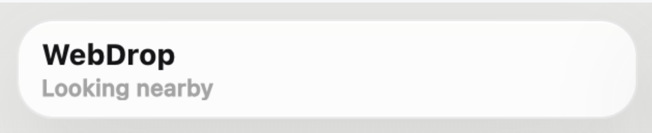


02

## WebDrop status pill

Shows the WebDrop label and nearby or connected status without exposing transfer controls too early.

03

## Settings icon

Opens the settings sheet for profile icon, ring color, language, app information, and version.

# Interface guide

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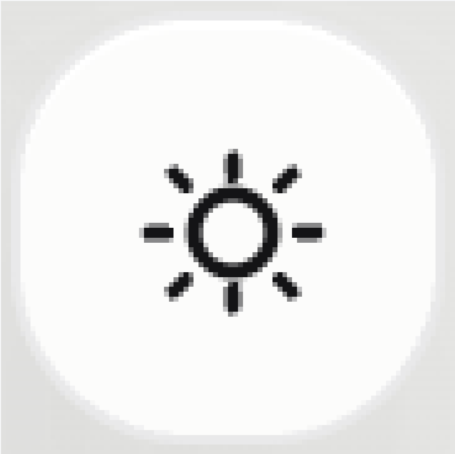
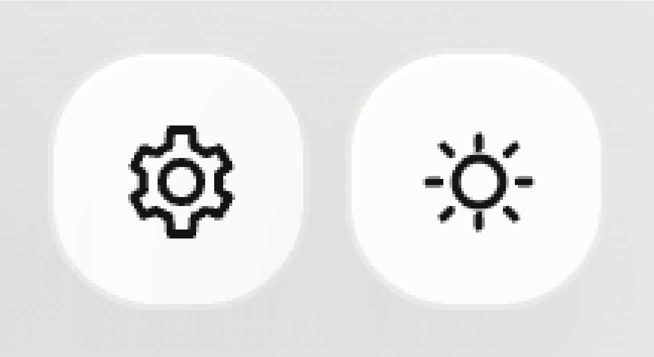


04

## Theme toggle

Switches between light and dark mode from the main screen.



05

## Header actions

Settings and theme controls are grouped in the top-right header area.

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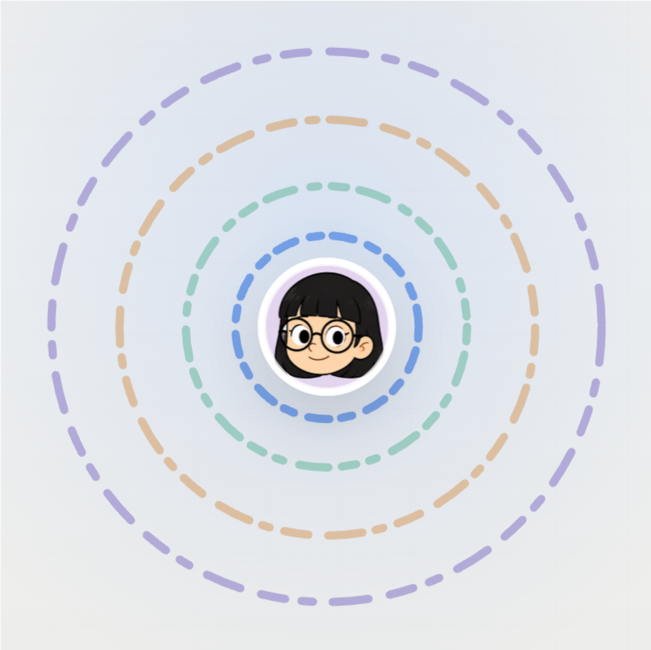
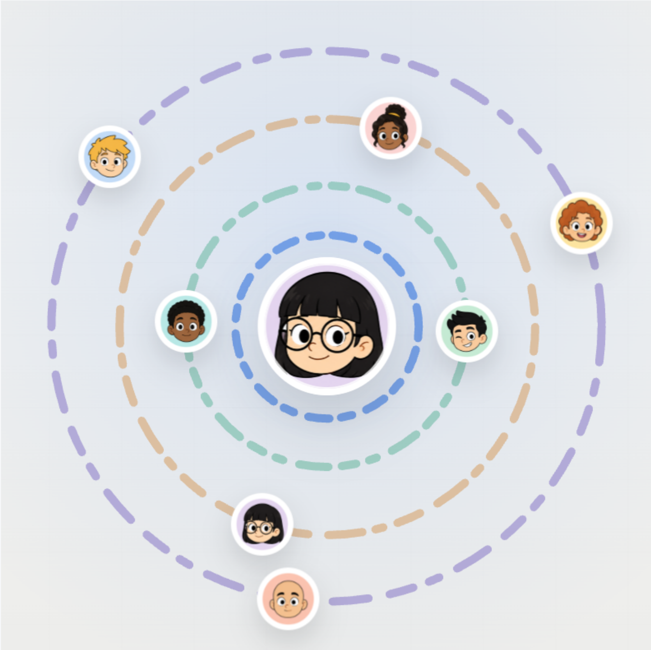


06

## Orbit layers without peers

The clean App Clip-inspired orbit pattern before showing nearby devices.



07

## Orbit layers with peers

Nearby candidates sit on separate rings with static profile icons so orbital movement remains calm.

# Interface guide

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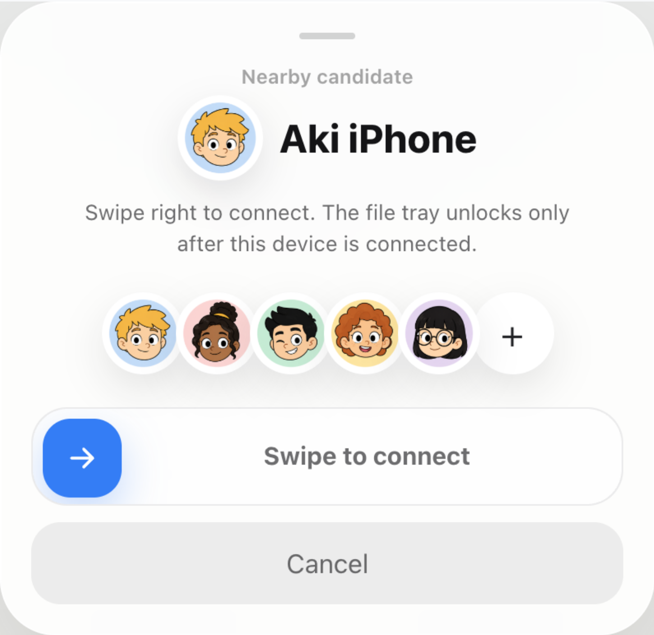
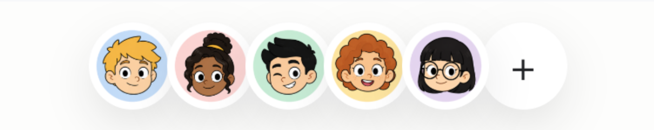



08

## Connect sheet

Peer selection opens a bottom sheet where the user confirms by swiping right.




09

## Friend strip

The candidate strip uses the same profile style as the orbit icons, with a small overlap and visible circular edges.

05 / UI CATALOG 06

# Interface guide

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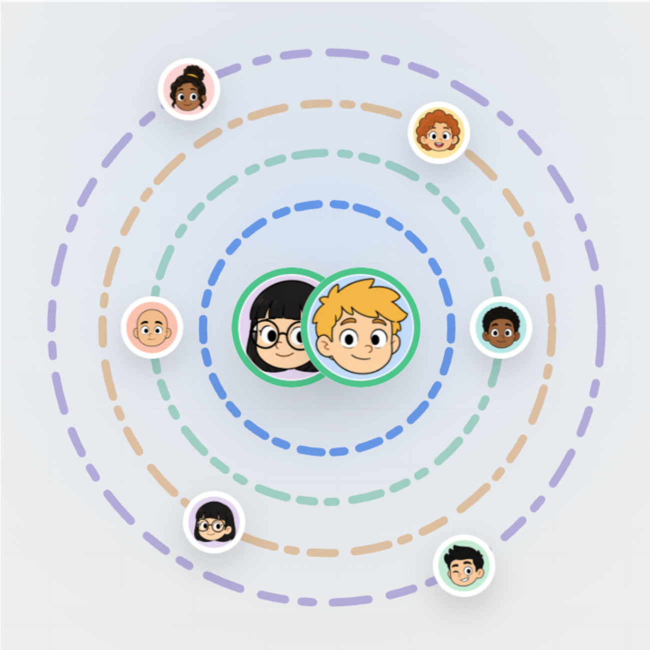

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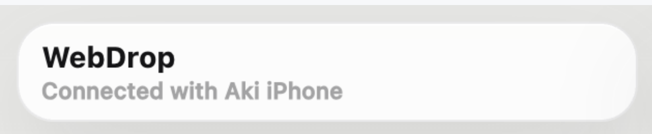
## Connected Venn state

The current device and selected peer overlap as an equal-size, static Venn-style pair with animated connected rings.

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## Connected status pill

After connection, the status line names the connected peer.

# Interface guide

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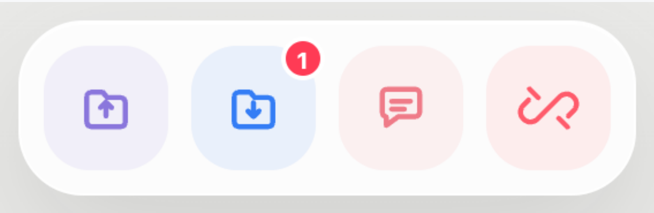

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## Connected dock

Send, receive, chat, and disconnect appear only after the connection is verified.

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## Send icon

Opens the send sheet.

# Interface guide

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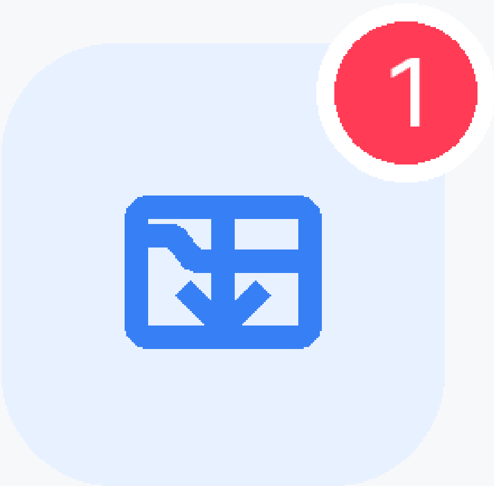


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## Receive icon

Opens received files and shows a badge when files are available.

15

## Chat icon

Opens the peer chat sheet.

# Interface guide

Approved WebDrop 1.0.10 UI inventory

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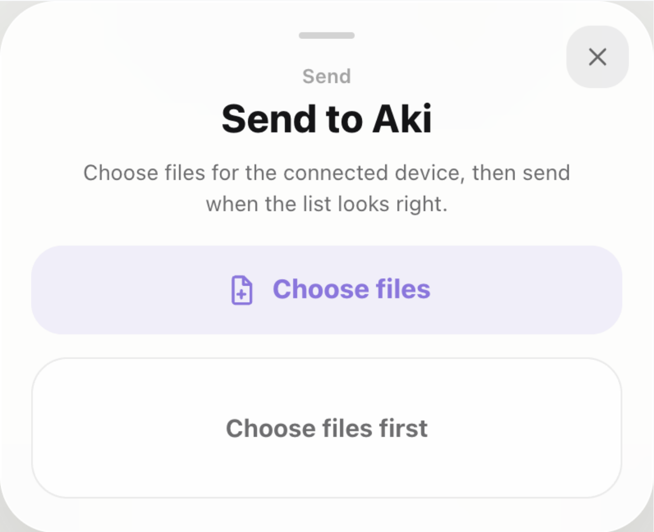
## Disconnect icon

Disconnects directly with a short release animation.

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## Send sheet before file selection

The user chooses files first; the send swipe remains disabled until a file is selected.

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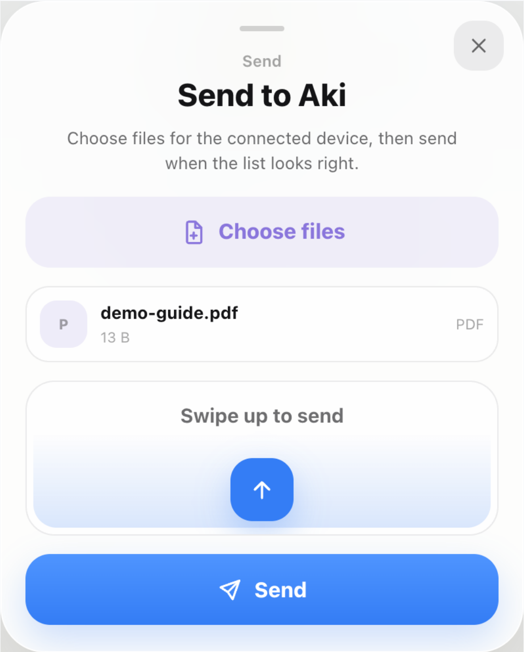
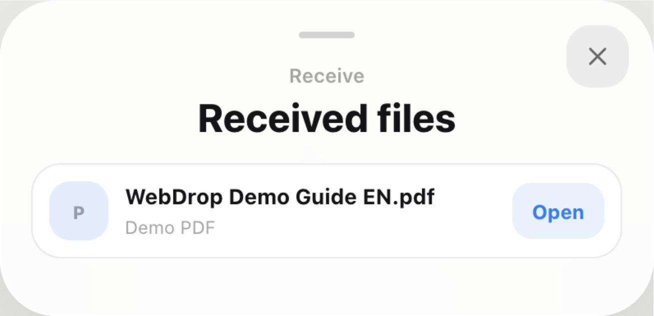



18

## Send sheet after file selection

A selected file appears in the list and the swipe-up send control keeps its icon below the text.

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## Receive sheet

Received demo PDFs appear here with an Open action and the dock badge reflects the count.

# Interface guide

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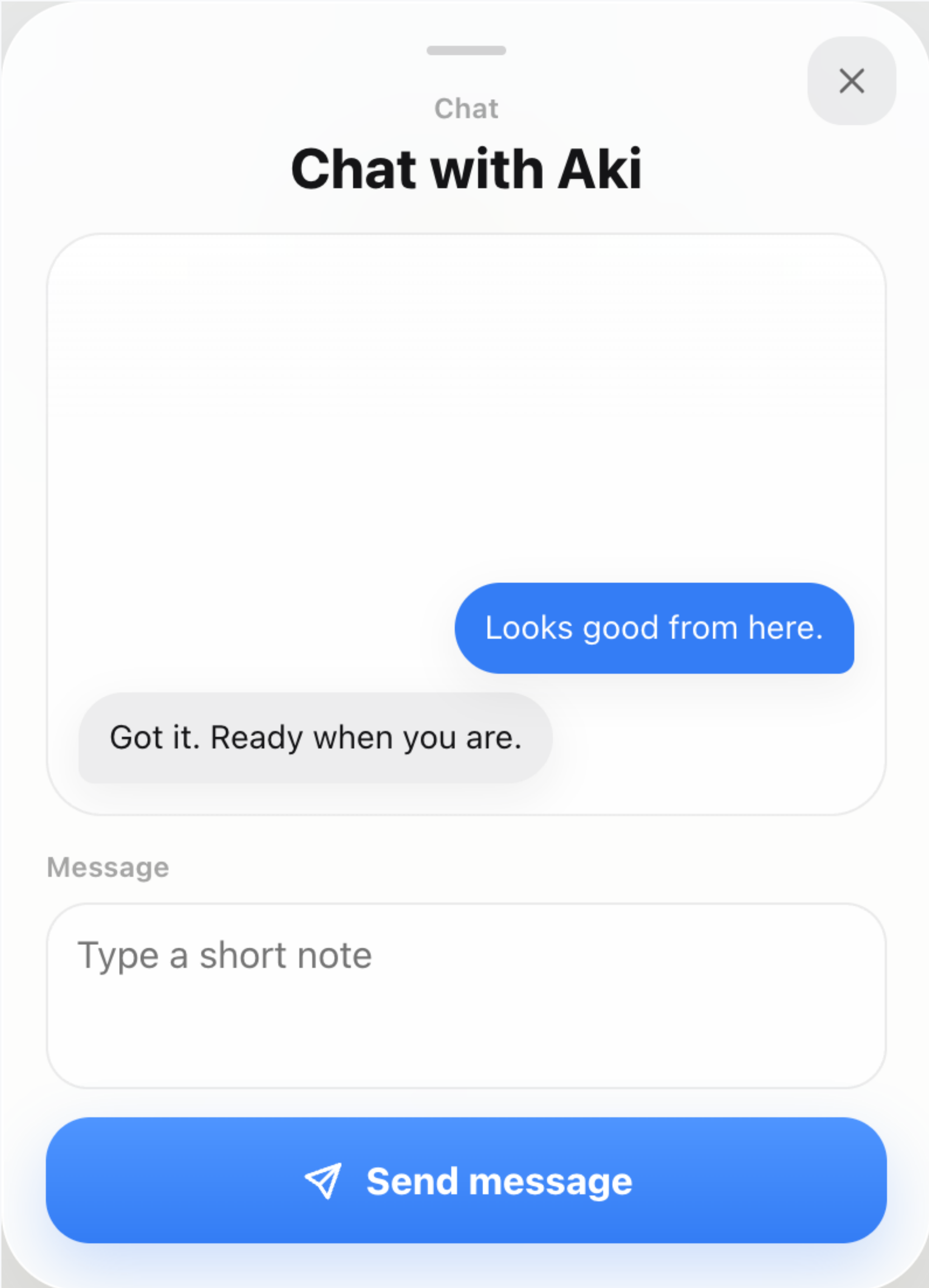


## Chat sheet

Chat is separated into its own scrollable bubble conversation instead of sharing the file tray.

# Interface guide

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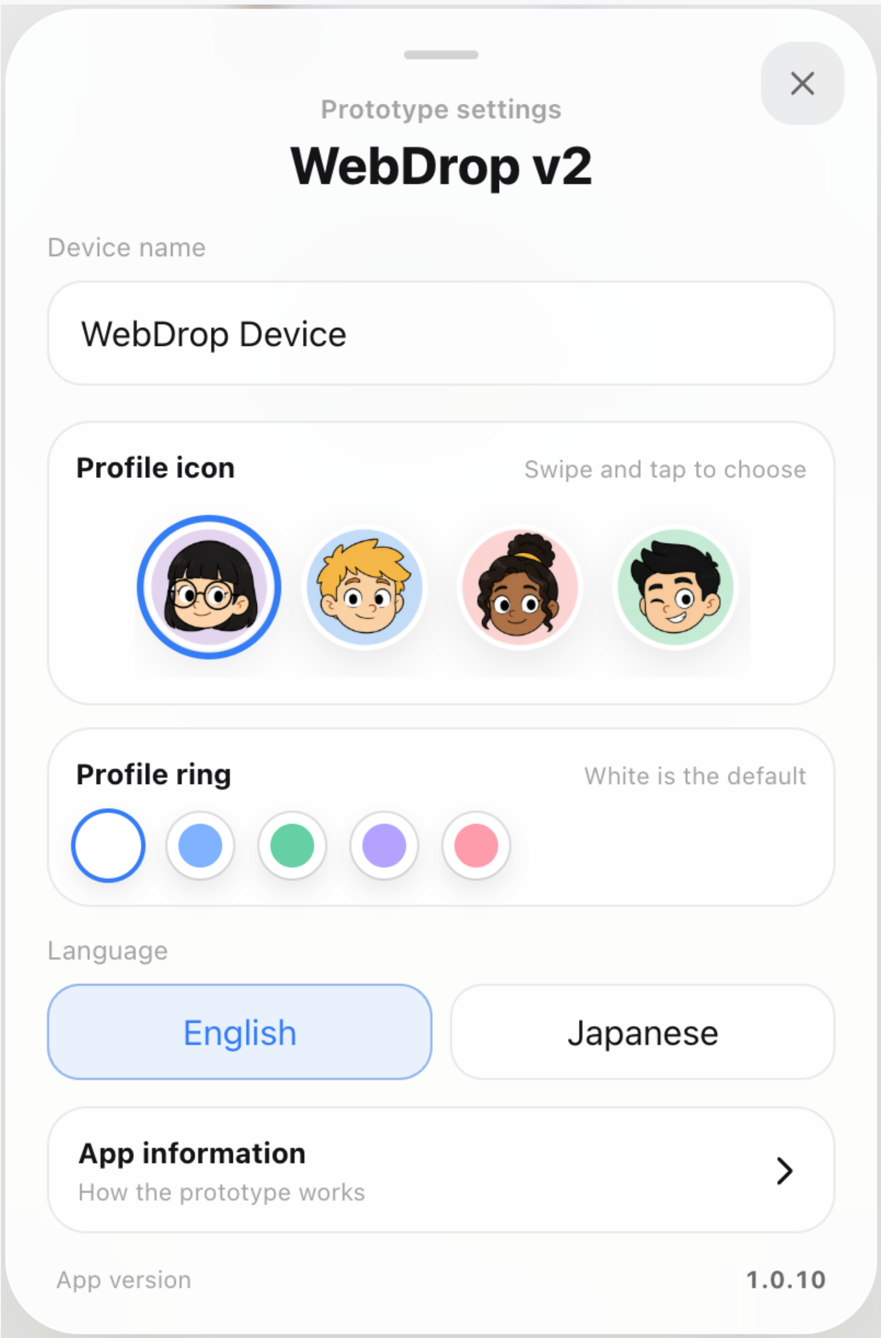


## Settings sheet

Settings hold device name, profile, ring, language, app information, and version.

# Interface guide

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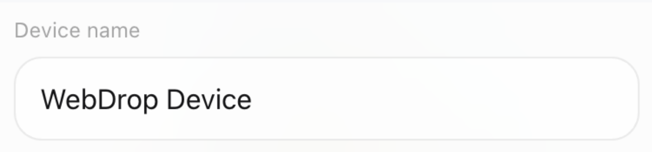


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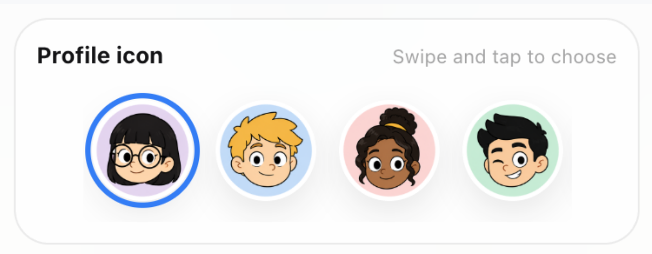
## Device name field

The device name can be edited, including clearing the final character without an unwanted reset.

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## Profile icon selector

Users swipe across the provided character icons instead of uploading a photo.

# Interface guide

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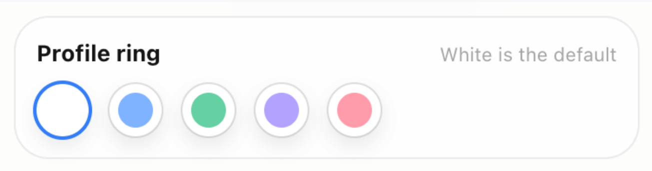


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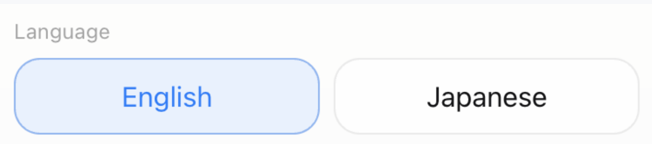
## Profile ring selector

The default ring is white, with optional blue, green, purple, and rose choices.

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## Language selector

Switches every app label between English and Japanese.

# Interface guide

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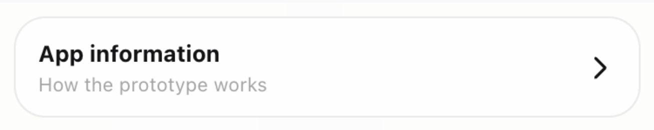


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## App information link

Moves design, stack, and orbit motion details into a separate sheet.


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## App version

Shows the current prototype version, 1.0.10, at the bottom of Settings.

# Interface guide

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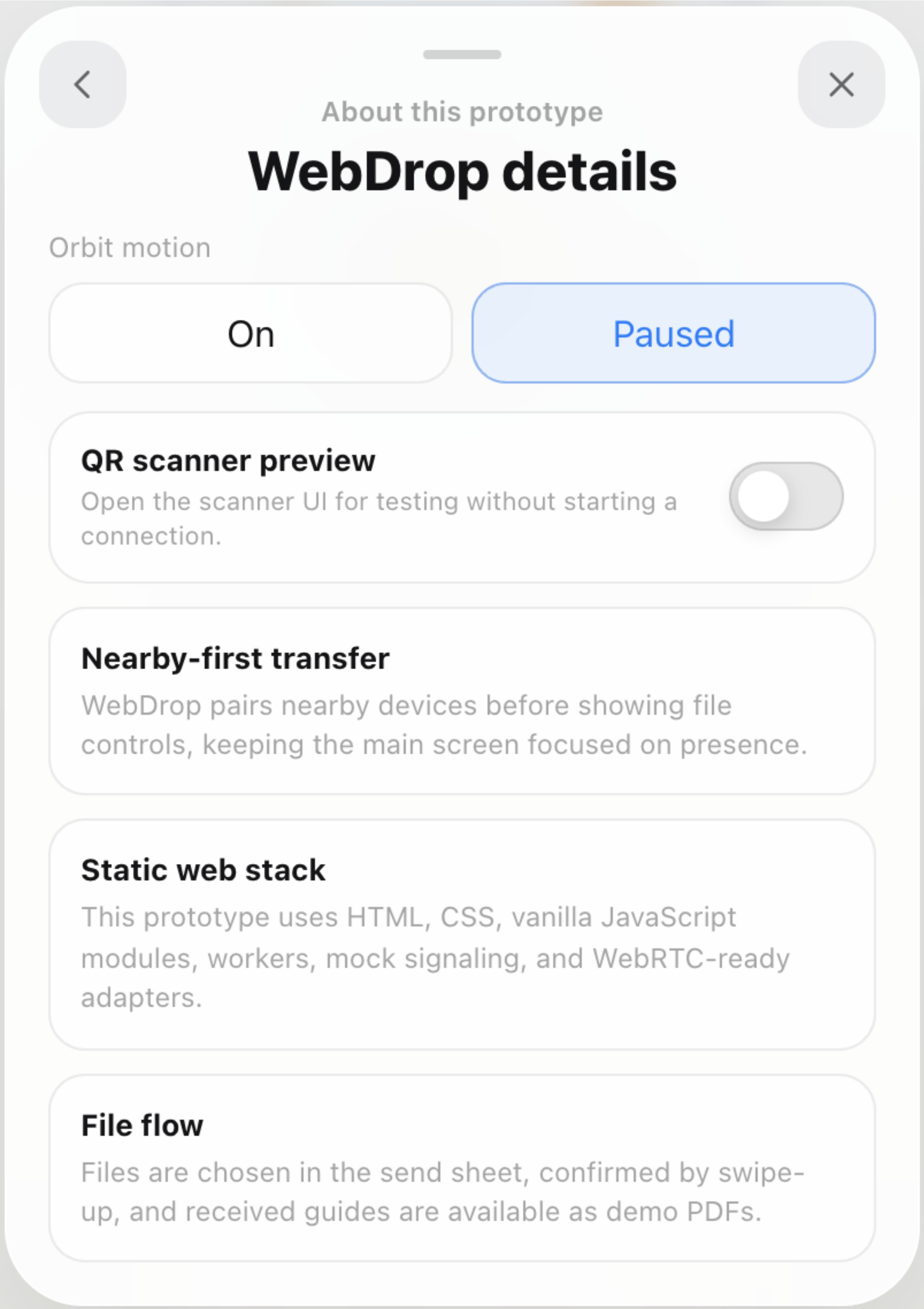


## App information sheet

Explains the prototype and contains orbit motion controls without cluttering the settings page.

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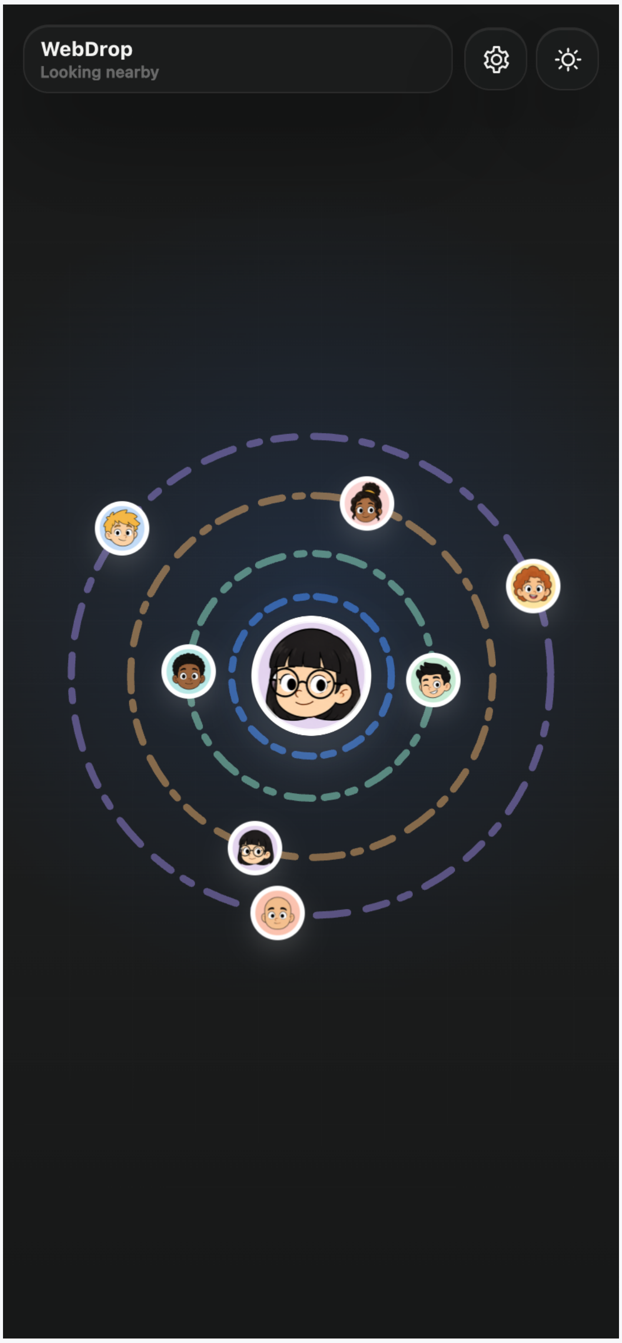


## Dark home screen

The same nearby-device UI in dark mode.

06 / NEXT

# Roadmap

**The remaining requirements are planned to be completed by the end of this week. From next week, the project focus moves to backend and server work.**

### Complete this week

Finish the remaining interface requirements, responsive verification, animation polish, accessibility checks, and documentation sign-off.

### Backend focus next week

Begin production WebSocket signaling, presence and invitation policy, session authentication, managed TURN integration, and operational limits.

### Production validation

Measure direct versus relay success, test cross-network transfers, verify storage recovery, and document realistic browser and file-size support.